AIMS 5710 Deep Learning Fundamentals and Theories
Homework 2

General Guidelines:

- Please check the submission deadline on **Blackboard** and submit your solutions via **Blackboard**.
- Each homework's deadline has a grace period of 2 hours.
- Each student has one chance of late submission within 12 hours of the deadline.
- All other late submissions will be given 0 points with no exception.
- **Do not** close your browser of app before you have successfully uploaded your files. It is your own responsibility of keeping your file integrity.

1. (30 points) **Back-propagation of convolution layers.** In lecture slides, we show that how to back-propagate derivatives from upper layer to bottom layer, i.e. from $\frac{\partial J}{\partial y}$ to $\frac{\partial J}{\partial x}$, and it can be achieved by rotating the kernel by 180° and performing convolution with the backward input as

$$\frac{\partial J}{\partial x} = \mathrm{rot}_{180^\circ}(w) * \frac{\partial J}{\partial y}$$

   However, this formula is derived for a single-channel input feature map $x \in \mathbb{R}^{h\times w\times 1}$, single-channel kernel $w \in \mathbb{R}^{k\times k\times 1}$, and a single-channel output feature map $y \in \mathbb{R}^{h\times w\times 1}$, where h, w are the height and width of the input feature map, and k is the size of the convolution kernel.

   Consider a more general case of multi-channel (c channels) input feature map $x \in \mathbb{R}^{h\times w\times c}$ and a multi-channel kernel $w \in \mathbb{R}^{k\times k\times c}$, which outputs a single-channel output feature map $y \in \mathbb{R}^{h\times w\times 1}$. Given the backward input $\frac{\partial J}{\partial y} \in \mathbb{R}^{h\times w\times 1}$, find the formula for $\frac{\partial J}{\partial x} \in \mathbb{R}^{h\times w\times c}$.

   You can use $w_1, w_2, \ldots, w_c$ to denote different channels of the multi-channel kernel w.

2. (30 points) **Linearity of convolution.** Explain with equations, why stacking two convolutional layers directly without non-linearity layers or functions separating them will result in a single convolutional layer.

3. (40 points) **Back-propgation of the BN layer.** Given the calculated mean and variance of a mini-batch, $\hat{\mu}_{\mathcal{B}}$ and $\hat{\sigma}^2_{\mathcal{B}}$, the foward inputs and outputs of a mini-batch are $\{x^{(1)} \in \mathbb{R}^n, \ldots, x^{(N)} \in \mathbb{R}^n\}$ and $\{y^{(1)} \in \mathbb{R}^n, \ldots, y^{(N)} \in \mathbb{R}^n\}$. Given the backward inputs $\{\frac{\partial J}{\partial y^{(1)}} \in \mathbb{R}^n, \ldots, \frac{\partial J}{\partial y^{(N)}} \in \mathbb{R}^n\}$, find $\frac{\partial J}{\partial x^{(i)}} \in \mathbb{R}^n$, $\frac{\partial J}{\partial \gamma} \in \mathbb{R}^n$, $\frac{\partial J}{\partial \beta} \in \mathbb{R}^n$.